

Web X.0 (IT5PC_CR3)

1.1 Teaching Scheme

Course Code	Course Name	Contact Hours	Credits
IT5PC_CR3	Web X.0	02	02

1.2 Examination and Evaluation Scheme

Formative Assessment		Summative Assessment				Total Marks
ISE		MSE		ESE		
Marks	Duration	Marks	Duration (Hrs.)	Marks	Duration (Hrs.)	
15	CA	15	1 Hour	45	2 Hours	75

1.3 Course Objectives

S. No.	Objectives
1	To familiarize students with web programming fundamentals and the digital evolution of web technologies.
2	To introduce students to JavaScript and TypeScript for building interactive and scalable web applications.
3	To enable students to create dynamic and responsive web applications using React and Angular frameworks based on the MVC architecture.
4	To introduce students to backend development using Node.js, Express, and Flask frameworks for building efficient and scalable web applications.
5	To build foundational knowledge of MongoDB, a document-oriented NoSQL database, and understand the concepts of Rich Internet Applications.

1.4 Course Outcomes

The students will be able to:	
CO1	Students will be able to <i>understand</i> the programming fundamentals and digital evolution of web Technology.
CO2	Students will be able to <i>apply</i> JavaScript for adding interactivity to web pages and <i>utilize</i> TypeScript for error-free code development.
CO3	Students will be able to <i>design</i> dynamic and responsive single-page front-end applications using React and AngularJS framework.
CO4	Students will be able to <i>build</i> back-end web applications using Flask, Node.js, and Express.
CO5	Students will be able to <i>Apply</i> MongoDB for front-end and back-end connectivity and <i>understand</i> the concepts of Rich Internet Applications (RIAs).

1.5 Course Contents

Module	Unit	Detailed Contents	Hours
		Prerequisite: HTML, CSS, Python Programming	
1.0		Fundamentals and Evolution of Web	05
	1.1	Working of web browser, HTTP/HTTPS, DNS, URL/URI, TLS, XML, JSON, DOM, REST API basics	
	1.2	Evolution from Web 1.0 to Web 5.0, Web analytics 2.0: Clickstream analysis, Strategy to choose your web analytics tool, Measuring the success of a website.	
	1.3	Semantic Web, Components, RDF, Ontology, SPARQL	
2.0		JavaScript to TypeScript Evolution	05
	2.1	Introduction to ES6 and differences from ES5, Variables, Conditions, Loops, Functions, Events, Arrow functions	
	2.2	DOM manipulation, Setting CSS using JavaScript, Classes, Inheritance, Iterators, Generators, Promises	
	2.3	Introduction to TypeScript and its architecture, TypeScript setup, Variables, Operators, Control structures, Functions, Classes, Objects, Modules in TypeScript	
3.0		Frontend Frameworks - React, Angular	05
	3.1	React installation, Folder and file structure, Components, Component lifecycle, State and Props, React router and Single page applications	
	3.2	Overview and need of AngularJS, Modules, Directives (Built-in & Custom), Expressions, Data binding, Controllers, Scope, Filters	
	3.3	AngularJS services, Routing using ng-Route, ng-Repeat, ng-style, ng-view	
4.0		Backend Frameworks – Node.js, Express.js, and Flask	06
	4.1	Node.js environment setup, First app, Asynchronous programming, Callback concept, Event loops, REPL, Buffers, Streams, File system	
	4.2	Introduction to Express.js, Express router, REST API, Generator, Authentication, Sessions	
	4.3	Flask environment setup, App routing, URL, Building Flask HTTP methods, Flask request object, Flask cookies	
5.0		MongoDB, REST APIs, and Rich Internet Applications	05
	5.1	Understanding MongoDB, MongoDB data types, Administering user accounts, Connecting to MongoDB from Node.js, Accessing and manipulating databases, Accessing MongoDB from Node.js, using Mongoose for structured schema and validation.	
	5.2	Overview of REST principles, CRUD operations, and integration with MongoDB using Express and Mongoose	
	5.3	Introduction to AJAX and its working, Overview of open-source frameworks like Django, Drupal, and Joomla	
		Total Hours	26

1.6 Suggested Learning Resources

1.6.1 Textbooks

1. Advanced Internet Technology, Dr. Deven Shah, DreamTech, 2014.

2. Rediscovering JavaScript: Master ES6, ES7, and ES8, Venkat Subramaniam, Pragmatic Bookshelf, 2018
3. Learning React Functional Web Development with React and Redux, Ist Ed., Alex Banks and Eve Porcello, O'Reilly, 2017
4. Flask Web Development: Developing Web Applications with Python, Ist Ed., Miguel Grinberg, O'Reilly, 2014
5. Getting MEAN with Mongo, Express, Angular, and Node, Simon Holmes, Clive Harber, IInd Ed., Manning Publications, 2019

1.6.2 Reference Books

1. TypeScript Quickly, Ist Ed. Yakov Fain and Anton Moiseev, Manning Publications, 2020.
2. Pro TypeScript: Application - Scale Javascript Development, IInd Ed., Steve Fenton, Apress, 2018
3. Node.js, MongoDB and Angular Web Development, IInd Ed., Brad Dayley, Brendan Dayley, Caleb Dayley, Pearson Education, 2018
4. MEAN Web Development, Amos Q. Haviv, Packt Publishing, 2014
5. Web Development with Node and Express, IInd Ed. Ethan Brown, O'Reilly, 2019

1.6.3 Web Resources

1. NPTEL course on Web Technology by Dr. Ashutosh Kumar Bhatt, Uttarakhand Open University, Haldwani
https://onlinecourses.swayam2.ac.in/e-learning/preview/nou25_cs09
2. Coursera course on Web Application Development with JavaScript and MongoDB by Dr Matthew Yee-King, University of London
<https://tinyurl.com/mpa2yfnt>
3. Udemy course on Node with React: Full Stack Web Development by Stephen Grider, Udemy course
<https://tinyurl.com/42fnk35h>